

Queensland University of Technology
Faculty of Science
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Computational Modelling of Multilateral Treaties

Stage 2

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Supervisory Team

Role	Name	Affiliation
Principal Supervisor	Prof. Michael Bode	School of Mathematical Sciences, QUT
Associate Supervisor	Prof. Jonathan Rhodes	School of Mathematical Sciences, QUT

Principal Supervisor — Prof. Michael Bode Prof. Bode is Deputy Director of Capability Development at SAEF (Securing Antarctica’s Environmental Future). His work centres on applied mathematics and conservation, with a focus on decision-making under uncertainty and mathematical biology. His previous work on Antarctic Treaty System (ATS) decision-making and conservation makes him an expert on consensus-based systems.

Associate Supervisor — Prof. Jonathan Rhodes Prof. Rhodes is Director of the Centre for Environment and Society. He specialises in spatial ecology, environmental governance, and the modelling of coupled human–natural systems, bridging quantitative models with real-world policy. His expertise gives him deep insight into the policy implications of environmental decisions, providing a grounding for the treaty-system applications explored in this research.

Chapter 1

Background and Literature Review

1.1 Introduction

Many international institutions use consensus or unanimity as the voting system choice. This can implicitly cap what these institutions can actually achieve. A consensus-based system earns broad legitimacy by granting equal veto power to every member. However, this gain in legitimacy trades off with slow adaptation, ultimately risking institutional inertia and decline. (Gray 2018). The Antarctic Treaty System (ATS) is one of those consensus-driven regimes, negotiated between a set of parties called consultative parties Secretariat of the Antarctic Treaty (2023). This group of countries, together with a second group lacking decision-making power (non-consultative parties), meets yearly at the Antarctic Treaty Consultative Meeting (ATCM). Before each meeting, consultative parties may submit Working Papers (WP) and Information Papers (IP) up to 45 days before the ATCM starts Secretariat of the Antarctic Treaty (2023). During ATCMs the parties discuss Working Papers and produce outcomes only by consensus. These outcomes take the form of Measures, Decisions, or Resolutions Antarctic Treaty Consultative Meeting (1995). As international relations strain and Antarctic environmental pressures intensify, the risk of losing the capacity to act increases. In 2022, for the first time, the ATCM parties failed to reach consensus on the final report; even as activity has increased, the regime has become less responsive to challenges Gardiner et al. (2025).

In a system based on consensus it is of utmost importance to understand why it succeeds or fails, and which parties drive it or block it, respectively. Existing literature has described the ATS by rates and diversity of outputs, and structurally by describing who co-authors with whom Larissa Lubiana Botelho et al. (2026) and Kumar et al. (2026). But there is no study where the reconstruction of a directional party intention was reconstructed from the textual records on each issue over time. Even further, that type

of measure has not been used to model the strategic behaviour that produces or breaks consensus. This thesis objective is to measure that and to build that model and asks if they contain the strategic intentions made by the consultative parties. It then asks whether these can model historical consensus dynamics.

The approach is a computational pipeline. First, it uses embedding techniques to discover topics, and NLI (Natural Language Inference) techniques for stance extraction. These would allow the construction of the preference measure. Second, it applies NLI to extract the opposition or aversion. It then tests the approach by comparing inputs (i.e. working papers) against observed outcomes. Finally, a Multi-Agent reinforcement learning is trained based on the previous outcomes to simulate the negotiation itself.

1.2 The Antarctic Treaty System as a consensus regime

In the ATS, Consultative Parties (CPs) reach outcomes exclusively by consensus. These discussions take place at the annual ATCM, where outcomes can take any of the following forms, per Secretariat of the Antarctic Treaty (2023, Rule 24):

- **Measures** are adopted texts containing provisions intended to become legally binding; these require approval by the relevant parties' legislative bodies.
- **Decisions** are adopted texts concerning internal organisational matters of the ATCM.
- **Resolutions** are non-binding recommendations or expressions of intent.

There are four types of input:

- **Working Papers** (Secretariat of the Antarctic Treaty (2023, Rule 48)): submitted by CPs and Observers, these require discussion at the ATCM and are the main source of outcomes. They are the only input from CPs that can produce an outcome; as such, they are the most likely place for each country to state its intentions.
- **Secretariat Papers** (Secretariat of the Antarctic Treaty (2023, Rule 49)): prepared and presented by the Secretariat, these produce organisational decisions as output, e.g. the budget.
- **Information Papers** (Secretariat of the Antarctic Treaty (2023, Rule 50)): supporting information for Working Papers or other discussions. They may be presented by CPs, Observers, Non-Consultative Parties (NCPs), and experts.

- **Background Papers** (Secretariat of the Antarctic Treaty (2023, Rule 51)): submitted by ATCM participants but not introduced during the meeting; they exist solely for the record.

Of all four, Working Papers are the only input uniquely submitted by CPs, requiring discussion and not confined to organisational matters. They are therefore the most likely place for countries to express their strategic intentions within this closed decision-making environment.

The consensus system makes the ATS a notable example of international cooperation; however, the same mechanism has become a major challenge as the number of CPs grows. Gardiner et al. (2025) show that the decision-making responsiveness of the ATS is declining; the unprecedented 2022 consensus failure is a clear example.

A notable case of contested consensus occurred in the late 1970s and 1980s, during the negotiations over mineral resources in Antarctica. The Convention on the Regulation of Antarctic Mineral Resource Activities (CRAMRA) was produced and then abandoned in favour of the Madrid Protocol (1991), which prohibited all mineral resource extraction in Antarctica (Jackson 2021). This well-documented episode provides a valuable canary test: a known case with documented positions against which any quantitative stance measure can be validated.

1.3 Quantitative and computational study of International institutions and treaties

In general the quantitative study of international institutions exist long before it was applied to Antarctica. Substantial literature maps the design of treaties and organisations into structured variables for example: their decision rules, dispute-settlement provisions, and enforcement mechanisms. Furthermore, these studies track these variable to to institutional performance and survival (Gray 2018). A more recent trend treats the *text* of international agreements and debates as data. They recover state positions from voting records and speeches (Baturu et al. 2017; Slapin and Proksch 2008). Both approaches show that preferences and design features of treaty regimes can be measured, not merely described.

The ATS has been analysed quantitatively in two ways: Gardiner et al. (2025) does performance accounts that measure the regime in aggregate, counting inputs and outputs, and the time legally binding measures take to enter into force. It concludes that responsiveness to challenges in the Antarctic region is declining. The second methodology is structural (Larissa Lubiana Botelho et al. (2026) and Kumar et al.

(2026)): mapping who collaborates with whom through co-authorship networks of ATCM documents. Both analyses are very valuable, and in this project both results are going to be used as an external check: the network structure as a replication target for the authorship layer, and the outcome series as a prediction target in Chapter 2. Neither, however, extracts topics from WPs in an unsupervised manner, nor yields a directional, country- and topic-level, time-resolved measure of policy preference.

1.4 Measuring policy preference from political text

Positional measures extracted from political texts are a well-developed field with two paradigms. The first paradigm is scaling and ideal-point methods, which recover a latent positional dimension without a pre-specified axis, i.e. there is no need for a regulate vs permissive comparison. These methods typically rely on word counts (i.e. frequencies) as their main underlying principle. This is the case of "Wordfish", introduced by Slapin and Proksch (2008) as an unsupervised positional scaling of party text. Lauderdale and Herzog (2016) addressed the problem that legislators talk about different things in different debates; they made a per-topic, per-debate scaling. On the other hand, Vafa et al. (2020) produced a probabilistic framework to place ideal points and the language of different actors. The closest dataset to the ATCM corpus was the one used by Baturu et al. (2017) to derive a state-level preference using United Nations (UN) debate speeches. Finally, a recent review by Parschan and Jakob (2025) condenses all these approaches in one paper. The second paradigm uses embeddings as its main tool. Rheault and Cochrane (2019) show that word-embedding geometry recovers ideological placement using parliamentary data. On the other hand, Rodriguez and Spirling (2022) provide a comprehensive analysis of what works and what does not when analysing political text using embedding techniques. They find embeddings a useful technique for those purposes but caution that the choice of embeddings strongly affects applied inference, and that the results must be validated. Finally, Licht (2023) and Chen et al. (2024) extend this approach to cross-lingual and multilingual settings.

Here, we propose to use both paradigms in the same pipeline but at different stages. This thesis differs from the ideal-point methods in an important way: ideal-point models recover a single stable latent position per actor, whereas the presented project requires a topic-conditioned stance on a given axis. The proposal in this thesis allows the parties to hold different positions on different topics through time. The topic structure is discovered using the embedding-based approach of Grootendorst (2022), which couples transformer representations with density clustering and class-based TF-IDF. The stance measure is then estimated with reference to the models of Burnham (2023) and Abercrombie and Batista-Navarro (2022). This thesis proposes to use discriminative transformer-NLI

models, which take a **premise** and a **hypothesis**. These NLI models output one of the prescribed classes and $P(\text{entailment} \mid \text{hypothesis})$, then the class with the highest probability is selected. The final results can be validated directly using the CRAMRA canary test (Jackson 2021), or indirectly by checking which countries cluster around (topic, stance) pairs and comparing this to the Larissa Lubiana Botelho et al. (2026) network results.

1.5 Modelling consensus, coalition formation and multi-actor decision-making

There two traditional group of models how groups reach consensus. The first is the most elegant but are the synchronization and opinion-dynamics models, like Kuramoto and opinion-changing-rate proposed by Leonardo L. Botelho (2022) which represent agreement as an coupled-oscillator convergence. These methods treat consensus as *convergence*. DeGroot (1974) average agents opinion iteratively using $x(t+1) = Wx(t)$ until they agree. There is also a bounded-version of this model developed by Hegselmann and Krause (2002). Another relevant model worth mentioning within this group is the stubborn agents (Friedkin and Johnsen 1990) where each agent does not fully update to match their neighbours, they stay, partially anchored to their initial opinion. The aforementioned group of models does not contain strategic preference, no veto and no learning, therefore, it cannot simulate the bargaining behaviour that occurs in consensus based systems.

The second group of models uses social choice and game theory to model consensus as a unanimity rule. Tsebelis (2002) develop a veto-players framework where every party has veto power, therefore, the status quo is only replaced by new measures when $u_i(p) \geq u_i(q)$ for every Party i . Consequently, when the number of parties increase and the spread of their position grows, the set of adoptable policies decreases and the systems stability increases. This conclusion is the same reached by Gardiner et al. (2025) when analysing the ATS qualitatively.

1.6 Negotiation and coalition formation simulation using reinforcement learning

The strategic view of consensus developed above implies that, in consensus-based systems, the best way to replicate behaviour among many heterogeneous Parties, each holding a veto and bargaining over a status quo, is to use repeated strategic interaction.

In the specific case of the quantitative simulation of the ATS over the years, its size and complexity mean that the closed-form equilibria of analytic game theory are intractable. In these cases, multi-agent reinforcement learning (MARL) is the appropriate tool, precisely because it keeps the game-theoretic substance while relaxing this constraint. Formally, MARL is defined over stochastic (Markov) games, a generalisation of the repeated and normal-form games of Section 1.5. Similarly, its learning procedures approximate equilibrium behaviour from interaction rather than solving for it in closed form (Littman 1994). In this context, neural networks (NN) enter only as function approximators that allow this learning to scale to large state and action spaces.

Foundational MARL work covers multi-issue bargaining (Chang 2021), negotiating-team formation (Bachrach et al. 2020), bargaining under risk and asymmetric information (Schmid and Zotero CH3] 2020), and search-improved game-theoretic learning (Li et al. 2023). In this field, there are review papers by Gronauer and Diepold (2022) on MARL and by Sarkar et al. (2022) on coalition formation. Most recently, MARL agents based on Large Language Models (LLMs) have been used to model political negotiation directly: for example, Moghimifar et al. (2024) simulate coalition formation in parliament using text data from party manifestos. Bolleddu (2025) present a MARL system for consensus building and negotiation based on synthetic data. Kulkarni et al. (2025) use negotiated agreements with game-theoretic reasoning to predict dynamic coalition formation in Diplomacy (a well-known benchmark for this type of simulation). Finally, some other benchmarks for real negotiation data are starting to appear (Benac et al. 2026; Gu et al. 2025). However, none of the works mentioned above link an empirically extracted and validated country-by-topic preference structure to a strategic simulation of a real consensus treaty regime.

1.7 Summary of Literature Gap

The above literature review has a common arc at every step of the process: it goes from qualitative methods, to descriptive measures, to structural mapping and, finally, to learning-based methods. This proposal is confined to the quantitative elements of that arc, each of which leaves questions unanswered. Furthermore, although this work is intended to develop a framework that could be extended to other international institutions, it will focus on the ATS. In this scenario, the descriptive work shows what the regime outputs but not who produces what. Structural and party-network analyses are undirected and unattributed. Text-based analysis recovers latent positions rather than topic-stance analysis through time. Finally, consensus has been simulated either as convergence or, where learning-based methods were used, on synthetic data or data other than the institution’s own deliberative records. No prior work extracts a validated,

signed, directional preference for each party over time from a consensus regime's own textual records. Not has any study used the extracted signed preferences per party to drive a strategic simulation in which agents have veto power and that is validated against realised outcomes. This research project intends to supply that coupling.

Chapter 2

Program and Design of the Research Investigation

2.1 Research Questions

The thesis addresses the gap mentioned above by answering these research questions:

Research Questions

- RQ1.** Can Parties' strategic positions be reconstructed from written records into a directional measure across topics and time?
- RQ2.** Does the reconstructed positions recover known history and predict outcomes?
- RQ3.** Can a multi-agent model parameterised by it reproduce consensus and test counterfactual decision rules?

2.2 Objectives, Methodology and Research Plan

2.2.1 Objective 1 / Chapter 1: The Support Layer

The **Aim** is to produce a defensible, directional measure of Parties' preferences over relevant topics through time, using documents the Parties submit themselves. For the ATCM case, Working Papers (WPs) are the input data because, of all submitted documents, these are the only ones submitted by Parties or Observers (non-Secretariat) that require discussion and action. This makes WPs the most likely text to contain Parties' strategic intentions. The analysis uses English-language papers, since these

represent the largest share of all WPs. It is also relevant that most of the available pre-trained embedding and NLI models were trained on English-language data.

The **approach** is a pipeline with multiple stages. The first set of stages applies BERTopic (Grootendorst 2022) to the whole corpus, producing a set of topics K with corresponding cluster centroids $\{\boldsymbol{\mu}_k\}_{k \in K}$, where each $\boldsymbol{\mu}_k \in \mathbb{R}^m$ in the m -dimensional embedding space. The second set of stages selects, for each document and topic, the passages relevant to that topic. Each document d is split into passages $P_d = \{p_1, \dots, p_{n_d}\}$, and each passage p is embedded as $\mathbf{e}_p \in \mathbb{R}^m$. Relevance acts as a gating mechanism that filters out irrelevant passages: for each document d and topic k , every passage $p \in P_d$ is scored by its cosine similarity to the topic centroid as follows,

$$\rho_{p,k} = \cos(\mathbf{e}_p, \boldsymbol{\mu}_k) = \frac{\mathbf{e}_p \cdot \boldsymbol{\mu}_k}{\|\mathbf{e}_p\| \|\boldsymbol{\mu}_k\|}.$$

The set of pairs (*document*, *topic*) is the top scoring pairs paragraphs above a threshold. This may result in several (*document*, *topic*) pairs for one document. Then looping over all these pairs, the stance on a named axis, is measured. These are then aggregated into a *Country* $\times$ *Topics* $\times$ *Year* tensor.

$$S_{c,k,y} = \sum_{d \in D(c,y)} w_{d,c} r_{d,k} s_{d,k} \quad (2.1)$$

Where $D(c, y)$ is the set of Working papers co-sponsored by country c in year y , $r_{d,k} \in [0, 1]$ is the relevance of document d to topic k , $s_{d,k} \in [-1, 1]$ is the signed stance of

2.3 Research Design Overview

The project is structured into three interconnected papers, each targeting one research question and one modelling contribution.

Table 2.1: Paper structure and milestones

Paper	Focus	Method	Target venue
1	Affinity dynamics	Network analysis, coalitional GT	TBD
2	Aversion dynamics	Veto-player model, ABM	TBD
3	Simulation framework	Agent-based model, RL	TBD

2.4 Paper 1 — Affinity Dynamics in Multilateral Treaties

2.5 Paper 2 — Aversion Dynamics and Consensus Breakdown

2.6 Paper 3 — Integrated Simulation Framework

2.7 Data Sources and Ethics

The primary data sources for this project are:

- Antarctic Treaty Consultative Meeting (ATCM) final reports and working papers (publicly available at <https://www.ats.aq/>).
- State-level foreign policy position datasets (e.g., UN voting records).
- Published network and voting data from the international relations literature.

All data are publicly available institutional records; no human-subjects research is involved and no ethics clearance is required beyond standard QUT research integrity protocols.

2.8 Timeline

Table 2.2: Indicative project timeline

Phase	Activity	Target Date
1	Literature review & theoretical framework	Month 6
2	Paper 1: data collection & analysis	Month 12
3	Paper 1: writing & submission	Month 15
4	Paper 2: model development & calibration	Month 18
5	Paper 2: writing & submission	Month 21
6	Paper 3: simulation framework development	Month 27
7	Paper 3: writing & submission	Month 30

Continued on next page

Phase	Activity	Target Date
8	Thesis integration & write-up	Month 36
9	Thesis submission	Month 38

2.9 Expected Contributions

1. A formal computational model of consensus dynamics in the ATS, the first of its kind at this level of institutional specificity.
2. A transferable agent-based simulation framework applicable to any consensus-based multilateral treaty.
3. New theoretical constructs (affinity/aversion indices) operationalising previously qualitative political science concepts.
4. Policy-relevant insights into treaty design features that improve stability under strategic pressure.

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Appendix A

Supplementary Material

A.1 Notation and Conventions

Table A.1: Notation used throughout the document

Symbol	Meaning
N	Set of treaty parties (agents), $ N = n$
$i, j \in N$	Individual state actors
$s_i \in S$	Strategy of party i
u_i	Utility function of party i
α_{ij}	Affinity score between parties i and j
β_{ij}	Aversion score between parties i and j
$\mathbf{W}$	DeGroot influence/weight matrix
$x_i(t)$	Position/opinion of party i at time t

A.2 Acronyms

ATS Antarctic Treaty System

ATCM Antarctic Treaty Consultative Meeting

ATP Antarctic Treaty Party

ABM Agent-Based Model

RL Reinforcement Learning

GT Game Theory

IR International Relations

QUT Queensland University of Technology

A.3 Candidate's Publication Record

No publications to date; in preparation.